

Hui (Hilla) Kuurila-Zhang

AI Research Scientist · Computer Vision · AI Safety · Deep Learning · LLM · GenAI · Robotics & Control

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EXPERIENCE

Doctoral Researcher — University of Oulu 2022–2026 · Oulu, Finland

- First author on four publications spanning adversarial training, diffusion model-based adversarial example generation, and the adversarial robustness evaluation of vision-language models.
- Designed an efficient adversarial training algorithm to improve deep learning models' robustness under worst-case, adversarial attacks. (BMCV 2023)
- Built a diffusion model-based unrestricted adversarial examples generation pipeline by steering pretrained latent diffusion models at inference time with adaptive, momentum-based adversarial guidance to produce photorealistic, semantically controlled adversarial images. (ICIP 2026)
- Ran adversarial robustness evaluations on open-source vision-language models, probing the failure modes and the transferability of the CLIP-based adversarial attack. (IEEE Intelligent Systems 2026)

Master's Thesis Worker — Volvo Cars (ADAS software team) 2021 (Jan–Oct) · Gothenburg, Sweden

- Built a deep convolutional neural network for object classification from low-cost ultrasonic sensor-array data, plus optimization algorithms for 3D object localization.
- Generated training data through COMSOL Multiphysics simulation.
- Worked in an agile/Scrum ADAS team with daily stand-ups.

EDUCATION

PhD, Computer Science — University of Oulu (Computer Vision & AI Safety) 2022–2026

MSc, Computer Science — Umeå University (Robotics & Control) 2019–2021

BSc, Physics — Umeå University (dual degree) 2018–2019

BE, Electronics Science & Technology — Shenzhen University (dual degree) 2015–2018

SELECTED PUBLICATIONS

Kuurila-Zhang, H., Wei, H., Chen, H., & Zhao, G. (2026). Beyond the ℓ_1 norm: a survey of unrestricted adversarial examples across methods, evaluation, and connections to mechanistic interpretability. Under review. (*first survey on unrestricted adversarial examples*)

Kuurila-Zhang, H., Chen, H., & Zhao, G. (2026). Venom: text-driven unrestricted adversarial example generation with diffusion models. IEEE ICIP.

Kuurila-Zhang, H., Chen, H., & Zhao, G. (2026). Evaluating the adversarial robustness of vision-language models for facial expression recognition. IEEE Intelligent Systems, 41(1), 105–112.

Kuurila-Zhang, H., Chen, H., & Zhao, G. (2023). Adaptive adversarial norm space for efficient adversarial training. BMVC.

CORE SKILLS

Machine learning: Deep learning theory · custom training routines & loss design · adversarial robustness & red-teaming · vision-language / diffusion / generative models · model evaluation

Robotics & control: Linear, adaptive, optimal/robust control · SLAM · path planning · ROS · MATLAB/Simulink · COMSOL Multiphysics

Embedded & edge: Embedded system design · motor control · analog/digital circuit design · microcontrollers (Teensy, Arduino Due) · sensor signal processing (LiDAR, ultrasonic)

Programming & tooling: Python, C++, Java, R, SQL, LaTeX · PyTorch, TensorFlow · Git · Linux · HPC/cluster computing · AWS

Languages: Chinese (native), English (full professional), Finnish (intermediate), Swedish (entry)